

Assessment Test: Software Tester (QA)

Duration: 01:15 Hrs

Full Name:	Aditya Chaudhary
Email ID:	chaudharyaadi1149@gmail.com
Contact Number:	8439609170

You are testing a **Task Management Application** with the following features:

- User **Registration & Login**
- Logged-in users can:
 - Create a task
 - View task list
 - Edit a task
 - Delete a task
- Tasks are stored in a database and displayed in a list

Assume the application is **close to production release** and has **no existing test documentation**.

Test Case Design

Write **test scenarios and test cases** for:

- Registration
- Login
- Task CRUD operations
- Input validation & error handling

Include **positive, negative, and edge cases**.

Bug Identification

List **8-10 potential bugs or risk areas** without executing the application. For each bug, mention:

- Bug description
- Severity (Critical / Major / Minor)
- Reason/Impact

Note

- You are **not expected to complete everything perfectly**.
- We evaluate **how you think, structure, and explain**, not just what you write.

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1 Test Case Design

1.1 Registration

ID	Scenario	Steps	Expected Result
TC-01	Verify if user can successfully register on the Application.	Enter Credentials and click login.	Successful Login
TC-02	Verify Credentials.	Enter credentials	Successfully verified
TC-03	Verify email format.	Enter email	Correct Email format
TC-04	Check for duplicate credentials	Enter duplicate credentials	Shows already registered
TC-06	Verify multiple authentication requests.	Try registering multiple time within a small time window	Successfully handles all requests and discards duplicate requests
TC-07	Check for SQL injection.	Enter SQL injection queries	Handles SQL injection and discards requests.
TC-08	Handle database failure during registration.	Register using credentials	Database handles request after restarting.
TC-09	Verify Special characters handling	Enter special characters in credentials	Handles special characters.

1.2 Login

ID	Scenario	Steps	Expected Result
TC-XX	Verify if user can login	Enter credentials on the login page	Successful Login
TC-XX	Login with incorrect credentials	Enter wrong password or username	Login Failed
TC-XX	Login with unregistered credentials	Enter unregistered credentials	Login failed shows not registered message.
TC-XX	Check login response time	Login using credentials	Successful login within expected time
TC-XX	Check for SQL injection in login page.	Enter SQL injection queries	Handles SQL injection and discards requests.
TC-XX	Handle database failure during login	Login using credentials (Database fails)	Database retrieves password successfully
TC-XX	Verify Special characters handling	Enter special characters in credentials	Handles special characters.
TC-XX	Verify Empty credentials	Leave credentials fields empty	Successful handling
TC-XX	Verify Max limit credentials	Enter credentials length up to allowed character limit.	Successful Login

1.3 Task CRUD operations

1.3.1 Task Creation

- Create tasks with valid data fields.
- Create tasks with minimum and maximum allowed data limit.
- Data submission with empty data fields.
- Verify special character and escape code handling.

- Create large amount of tasks within small amount of time.
- Verify duplicate task creation and handling.
- Verify task creation failure.
- Verify task is added successfully in the database.
- Handle multiple task creation requests in the database.
- Handle database failure during task creation.

1.3.1.1 Test Cases

- Create task with valid title and description data.
- Create task with title only.
- Create task with description only.
- Create task with empty title and description.
- Enter special characters.
- Check created task in the task list.

1.3.2 Task List

- Verify newly created task appear within expected time window.
- Verify if duplicate tasks appear in the list.
- Verify if all user task are listed.
- Verify task appear in their respective status section(Completed, Working, Halted etc.).
- Verify task list auto-refreshes after task is created, edited, or deleted.
- Verify list behaviour when list is empty.
- Verify task is accessible in the database.
- Handle database failure during task retrieval.

1.3.2.1 Test Cases

- List behaviour with zero tasks.
- List behaviour with one or more tasks.
- List behaviour when task is created.
- List behaviour when task is deletion.
- List behaviour when task is edited.
- Refresh task list.
- List behaviour when database not available.

1.3.3 Task Editing

- Verify task editing thorough valid data.
- Edit only single task field.
- Enter valid and invalid data.
- Edit tasks with minimum and maximum allowed data limit.
- Verify updated task appear in the list successfully.
- Verify task is updated successfully in the database.
- Handle multiple task update requests in the database.
- Handle database failure during editing.

1.3.3.1 Test Cases

- Edit task with valid title and description data.
- Edit task with title only.
- Edit task with description only.
- Remove a task title and description.
- Add special characters in the task's title and description.
- Check edited task in the task list.
- Verify task updated in the database.

1.3.4 Task Deletion

- Verify task deletion.
- Verify multiple task deletion requests for a single task.
- Verify deleted task is removed from the list successfully.
- Verify deletion confirmation behaviour.
- Verify deletion cancellation behaviour.
- Verify multiple deletion requests for the same task.
- Handle multiple deletion requests in the database.
- Handle database failure during deletion.

1.3.4.1 Test Cases

- Delete existing task.
- Confirm deletion behaviour.
- Cancel Deletion behaviour.
- Multiple deletion requests for the same task.
- Delete already deleted task.

1.4 Input Validation

ID	Scenario	Steps	Expected Result
TC-XX	Empty required credential field	Leave credentials field empty	Login failed and error message is shown
TC-XX	Enter invalid email	Enter email without “@” and in other invalid formats	Email is rejected
TC-XX	Enter values at minimum allowed length	Enter values length at minimum allowed length	Successful Login/Registration
TC-XX	Password/Username mismatch	Enter incorrect password and username	Login/Registration Failure and error is displayed to user.
TC-XX	Enter values at maximum allowed length	Enter values length at minimum allowed length	Successful Login/Registration
TC-XX	Enter values exceeding maximum allowed length	Enter values greater than maximum allowed length	Registration/Login failed
TC-XX	Enter values containing special characters	Enter special values in input fields	Special characters are handled successfully and Login/Registration successfully

1.5 Error Handling

ID	Scenario	Steps	Expected Result
TC-XX	Password/Username mismatch	Enter incorrect password and username	Login/Registration Failure and error is displayed to user.
TC-XX	Database Failure	Database fails during mid user session	User data is successfully committed and preserved.
TC-XX	User Network Failure	User Networks fails mid request.	User request is halted and Session is stopped
TC-XX	Server Network Failure	Server Fails to connect to the Network	Data is committed and preserved, server tries to reconnect to the Network.
TC-XX	Task Creation Failure	Task creation fails on the server when creating a task	Error message is shown and task creation is retired.

ID	Scenario	Steps	Expected Result
TC-XX	Task Update Failure	Task Update fails on the server when editing a task	Error message is shown and task is updated later.
TC-XX	Task Deletion Failure	Task deletion fails on the server during task deletion	Error message is shows and task deletion is retired
TC-XX	Unauthorized task access	Unauthorized user tries to access another user task list	Unauthorized user is denied to access any task

2 Bug Identification

ID	Bug / Risk Description	Severity	Reason / Impact
1	User can access task list with logging in	Critical	Security Risk, A User may gain access to application data with authentication.
2	One user can view and edit another user task list	Critical	Security Risk, User data loss, Unauthorized user can access and edit another user tasks violating data isolation.
3	Credentials are stored and transferred in plain text.	Critical	Security Risk, Users credentials and vital information is compromised.
4	Duplicate task creation when multiple requests send quickly	Critical	Bad User Experience, User list may get filled with a lot of duplicate tasks ruining user experience.
5	Deleting task respectively also deletes other task	Critical	User data loss, User's important or required task may get deleted.
6	Deleted task appear in the task list	Critical	Bad User experience, unwanted task still appear in the task list.
7	Input fields accepts invalid or excessively large length input	Major	Bad user experience/Data corruption, User data may not be stored correctly in the database or it might break the frontend UI.
8	Edited tasks are not saved to the database and reverted after refresh	Major	Bad User Experience
9	SQL injection is not handled properly	Critical	Security Risk, Could provide Unauthorized person access to the database and Critical user information.
10	Long text leaks out or text containers	Minor	Bad User Experience